Program Summary - Assignment 1.sas

Execution Environment

Author:	u64425566
File:	/home/u64425566/BAN_100/Assignment 1.sas
SAS Platform:	Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64
SAS Host:	ODAWS02-USW2-2.ODA.SAS.COM
SAS Version:	9.04.01M8P02222023
SAS Locale:	en_US
Submission Time:	4/17/2026, 12:33:08 PM
Browser Host:	BRAS-BASE-TOROON5049W-GRC-02-142-126-227-211.DSL.BELL.CA
User Agent:	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 Edg/147.0.0.0
Application Server:	ODAMID00-USW2-2.ODA.SAS.COM

Code: Assignment 1.sas

```
/*----------------------------------------------------*/
/* Cooperative vs Competitive Classroom Activities  */
/* Descriptive Statistics Analysis                  */
/*----------------------------------------------------*/

/* Step 1: Create the dataset */
data classroom;
    input cooperation competition;
    datalines;
70 66
70 62
71 68
72 68
72 75
72 74
73 80
73 80
74 82
84 85
80 85
79 86
80 88
85 90
;
run;

/* Step 2: Display the dataset */
proc print data=classroom;
    title "Classroom Data: Cooperation vs Competition";
run;
```

```
/*-----------------------------------------------*/
/* (a) Measures of Central Tendency              */
/* Mean, Median, Mode                            */
/*-----------------------------------------------*/
proc means data=classroom mean median mode maxdec=2;
    var cooperation competition;
    title "Measures of Central Tendency";
run;

/*-----------------------------------------------*/
/* (b) Measures of Variation                     */
/* Range, Variance, Standard Deviation           */
/*-----------------------------------------------*/
proc means data=classroom range var std maxdec=2;
    var cooperation competition;
    title "Measures of Variation";
run;

/*-----------------------------------------------*/
/* (c) Measures of Shape                         */
/* Skewness and Kurtosis                         */
/*-----------------------------------------------*/
proc means data=classroom skewness kurtosis maxdec=2;
    var cooperation competition;
    title "Measures of Shape";
run;

/*-----------------------------------------------*/
/* Optional: Detailed Distribution Analysis      */
/*-----------------------------------------------*/
proc univariate data=classroom;
    var cooperation competition;
    title "Detailed Univariate Analysis";
run;




/*-----------------------------------------------*/
/* Baby Birth Weight Analysis                    */
/* Investigating Relationship Using T-Tests     */
/*-----------------------------------------------*/

/* Step 1: Inspect the data */
proc import out= babyweight
datafile= "/home/u64425566/BAN_100/File BIRTH(1).xlsx"
dbms= xlsx;
getnames= yes;
run;
```

```
proc print data=babyweight(obs=10);
    title "Sample of Baby Birth Weight Data";
run;

/*-----------------------------------------------------*/
/* Step 2: Data audit – missing values                 */
/*-----------------------------------------------------*/
proc means data=babyweight n nmiss;
    var Weight;
    title "Missing Value Check for Birth Weight";
run;

/*-----------------------------------------------------*/
/* Step 3: Descriptive statistics                      */
/*-----------------------------------------------------*/
proc means data=babyweight mean std min max;
    var Weight;
    title "Descriptive Statistics for Birth Weight";
run;

/*-----------------------------------------------------*/
/* Step 4: Distribution check (Normality)              */
/*-----------------------------------------------------*/
proc univariate data=babyweight;
    var Weight;
    histogram Weight / normal;
    title "Distribution of Infant Birth Weight";
run;

/*-----------------------------------------------------*/
/* Step 5: T-Test by Baby Sex                          */
/*-----------------------------------------------------*/
proc ttest data=babyweight;
    class Boy;
    var Weight;
    title "T-Test: Birth Weight by Baby Sex";
run;

/*-----------------------------------------------------*/
/* Step 6: T-Test by Maternal Smoking Status           */
/*-----------------------------------------------------*/
proc ttest data=babyweight;
    class MomSmoke;
    var Weight;
    title "T-Test: Birth Weight by Maternal Smoking";
run;

/*-----------------------------------------------------*/
/* Step 7: T-Test by Marital Status                    */
/*-----------------------------------------------------*/
proc ttest data=babyweight;
```

```
    class Married;
    var Weight;
    title "T-Test: Birth Weight by Marital Status";
run;
```

Log: Assignment 1.sas

Notes (24)

```
1          OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
68
69         /*----------------------------------------------------*/
70         /* Cooperative vs Competitive Classroom Activities  */
71         /* Descriptive Statistics Analysis                  */
72         /*----------------------------------------------------*/
73
74         /* Step 1: Create the dataset */
75         data classroom;
76             input cooperation competition;
77             datalines;

NOTE: The data set WORK.CLASSROOM has 14 observations and 2 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              667.25k
      OS Memory           19620.00k
      Timestamp           04/17/2026 04:33:01 PM
      Step Count                        24  Switch Count  2
      Page Faults                       0
      Page Reclaims                     137
      Page Swaps                        0
      Voluntary Context Switches        10
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264


92         ;
93         run;
94
95         /* Step 2: Display the dataset */
96         proc print data=classroom;
97             title "Classroom Data: Cooperation vs Competition";
98         run;

NOTE: There were 14 observations read from the data set WORK.CLASSROOM.
NOTE: PROCEDURE PRINT used (Total process time):
      real time           0.01 seconds
      user cpu time       0.01 seconds
      system cpu time     0.00 seconds
      memory              1340.46k
      OS Memory           20644.00k
      Timestamp           04/17/2026 04:33:01 PM
      Step Count                        25  Switch Count  1
```

```
      Page Faults                       0
      Page Reclaims                     314
      Page Swaps                        0
      Voluntary Context Switches        8
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           8


99
100       /*-----------------------------------------------*/
101       /* (a) Measures of Central Tendency              */
102       /* Mean, Median, Mode                            */
103       /*-----------------------------------------------*/
104       proc means data=classroom mean median mode maxdec=2;
105           var cooperation competition;
106           title "Measures of Central Tendency";
107       run;

NOTE: There were 14 observations read from the data set WORK.CLASSROOM.
NOTE: PROCEDURE MEANS used (Total process time):
      real time           0.02 seconds
      user cpu time       0.03 seconds
      system cpu time     0.01 seconds
      memory              7164.59k
      OS Memory           27064.00k
      Timestamp           04/17/2026 04:33:01 PM
      Step Count                        26  Switch Count  2
      Page Faults                       0
      Page Reclaims                     1733
      Page Swaps                        0
      Voluntary Context Switches        28
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           0


108
109       /*-----------------------------------------------*/
110       /* (b) Measures of Variation                     */
111       /* Range, Variance, Standard Deviation           */
112       /*-----------------------------------------------*/
113       proc means data=classroom range var std maxdec=2;
114           var cooperation competition;
115           title "Measures of Variation";
116       run;

NOTE: There were 14 observations read from the data set WORK.CLASSROOM.
NOTE: PROCEDURE MEANS used (Total process time):
      real time           0.01 seconds
      user cpu time       0.01 seconds
      system cpu time     0.00 seconds
      memory              6704.62k
      OS Memory           27320.00k
      Timestamp           04/17/2026 04:33:01 PM
      Step Count                        27  Switch Count  2
      Page Faults                       0
```

```
      Page Reclaims                     1467
      Page Swaps                        0
      Voluntary Context Switches        27
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           0


117
118         /*-----------------------------------------------*/
119         /* (c) Measures of Shape                          */
120         /* Skewness and Kurtosis                          */
121         /*-----------------------------------------------*/
122         proc means data=classroom skewness kurtosis maxdec=2;
123             var cooperation competition;
124             title "Measures of Shape";
125         run;

NOTE: There were 14 observations read from the data set WORK.CLASSROOM.
NOTE: PROCEDURE MEANS used (Total process time):
      real time           0.01 seconds
      user cpu time       0.01 seconds
      system cpu time     0.00 seconds
      memory              6698.68k
      OS Memory           27320.00k
      Timestamp           04/17/2026 04:33:01 PM
      Step Count                        28  Switch Count  2
      Page Faults                       0
      Page Reclaims                     1394
      Page Swaps                        0
      Voluntary Context Switches        27
      Involuntary Context Switches      1
      Block Input Operations            0
      Block Output Operations           0


126
127         /*-----------------------------------------------*/
128         /* Optional: Detailed Distribution Analysis       */
129         /*-----------------------------------------------*/
130         proc univariate data=classroom;
131             var cooperation competition;
132             title "Detailed Univariate Analysis";
133         run;

NOTE: PROCEDURE UNIVARIATE used (Total process time):
      real time           0.06 seconds
      user cpu time       0.06 seconds
      system cpu time     0.00 seconds
      memory              1014.34k
      OS Memory           22436.00k
      Timestamp           04/17/2026 04:33:01 PM
      Step Count                        29  Switch Count  1
      Page Faults                       0
      Page Reclaims                     153
      Page Swaps                        0
      Voluntary Context Switches        13
```

```
      Involuntary Context Switches      2
      Block Input Operations            0
      Block Output Operations           48



      /*-----------------------------------------------*/
      /* Baby Birth Weight Analysis                     */
      /* Investigating Relationship Using T-Tests       */
      /*-----------------------------------------------*/

      /* Step 1: Inspect the data */
      proc import out= babyweight
      datafile= "/home/u64425566/BAN_100/File BIRTH(1).xlsx"
      dbms= xlsx;
      getnames= yes;
      run;

NOTE: The import data set has 50000 observations and 10 variables.
NOTE: WORK.BABYWEIGHT data set was successfully created.
NOTE: PROCEDURE IMPORT used (Total process time):
      real time           3.61 seconds
      user cpu time       3.58 seconds
      system cpu time     0.01 seconds
      memory              4647.40k
      OS Memory           27128.00k
      Timestamp           04/17/2026 04:33:05 PM
      Step Count                        30  Switch Count  2
      Page Faults                       0
      Page Reclaims                     1045
      Page Swaps                        0
      Voluntary Context Switches        26
      Involuntary Context Switches      8
      Block Input Operations            0
      Block Output Operations           7944


      proc print data=babyweight(obs=10);
          title "Sample of Baby Birth Weight Data";
      run;

NOTE: There were 10 observations read from the data set WORK.BABYWEIGHT.
NOTE: PROCEDURE PRINT used (Total process time):
      real time           0.01 seconds
      user cpu time       0.02 seconds
      system cpu time     0.00 seconds
      memory              2028.59k
      OS Memory           24740.00k
      Timestamp           04/17/2026 04:33:05 PM
      Step Count                        31  Switch Count  1
      Page Faults                       0
      Page Reclaims                     257
      Page Swaps                        0
```

```
      Voluntary Context Switches        8
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           16


153
154        /*----------------------------------------------------*/
155        /* Step 2: Data audit – missing values                */
156        /*----------------------------------------------------*/
157        proc means data=babyweight n nmiss;
158            var Weight;
159            title "Missing Value Check for Birth Weight";
160        run;

NOTE: There were 50000 observations read from the data set WORK.BABYWEIGHT.
NOTE: PROCEDURE MEANS used (Total process time):
      real time           0.01 seconds
      user cpu time       0.01 seconds
      system cpu time     0.01 seconds
      memory              8199.21k
      OS Memory           30920.00k
      Timestamp           04/17/2026 04:33:05 PM
      Step Count                        32  Switch Count  2
      Page Faults                       0
      Page Reclaims                     1847
      Page Swaps                        0
      Voluntary Context Switches        37
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           0


161
162        /*----------------------------------------------------*/
163        /* Step 3: Descriptive statistics                     */
164        /*----------------------------------------------------*/
165        proc means data=babyweight mean std min max;
166            var Weight;
167            title "Descriptive Statistics for Birth Weight";
168        run;

NOTE: There were 50000 observations read from the data set WORK.BABYWEIGHT.
NOTE: PROCEDURE MEANS used (Total process time):
      real time           0.01 seconds
      user cpu time       0.02 seconds
      system cpu time     0.00 seconds
      memory              8202.21k
      OS Memory           30920.00k
      Timestamp           04/17/2026 04:33:05 PM
      Step Count                        33  Switch Count  2
      Page Faults                       0
      Page Reclaims                     1845
      Page Swaps                        0
      Voluntary Context Switches        37
      Involuntary Context Switches      0
      Block Input Operations            0
```

```
      Block Output Operations           0


169
170        /*----------------------------------------------------*/
171        /* Step 4: Distribution check (Normality)             */
172        /*----------------------------------------------------*/
173        proc univariate data=babyweight;
174            var Weight;
175            histogram Weight / normal;
176            title "Distribution of Infant Birth Weight";
177        run;

NOTE: PROCEDURE UNIVARIATE used (Total process time):
      real time           2.24 seconds
      user cpu time       0.15 seconds
      system cpu time     0.01 seconds
      memory              15165.92k
      OS Memory           37120.00k
      Timestamp           04/17/2026 04:33:07 PM
      Step Count                        34  Switch Count  1
      Page Faults                       0
      Page Reclaims                     3830
      Page Swaps                        0
      Voluntary Context Switches        989
      Involuntary Context Switches      7
      Block Input Operations            0
      Block Output Operations           512


178
179        /*----------------------------------------------------*/
180        /* Step 5: T-Test by Baby Sex                         */
181        /*----------------------------------------------------*/
182        proc ttest data=babyweight;
183            class Boy;
184            var Weight;
185            title "T-Test: Birth Weight by Baby Sex";
186        run;

NOTE: PROCEDURE TTEST used (Total process time):
      real time           1.56 seconds
      user cpu time       0.86 seconds
      system cpu time     0.19 seconds
      memory              18047.59k
      OS Memory           50828.00k
      Timestamp           04/17/2026 04:33:09 PM
      Step Count                        35  Switch Count  123
      Page Faults                       0
      Page Reclaims                     52115
      Page Swaps                        0
      Voluntary Context Switches        9686
      Involuntary Context Switches      35
      Block Input Operations            0
      Block Output Operations           90008
```

```
187
188         /*----------------------------------------------------*/
189         /* Step 6: T-Test by Maternal Smoking Status         */
190         /*----------------------------------------------------*/
191         proc ttest data=babyweight;
192             class MomSmoke;
193             var Weight;
194             title "T-Test: Birth Weight by Maternal Smoking";
195         run;
```

```
NOTE: PROCEDURE TTEST used (Total process time):
      real time           1.66 seconds
      user cpu time       1.09 seconds
      system cpu time     0.22 seconds
      memory              17143.81k
      OS Memory           50676.00k
      Timestamp           04/17/2026 04:33:10 PM
      Step Count                        36  Switch Count  117
      Page Faults                       0
      Page Reclaims                     52411
      Page Swaps                        0
      Voluntary Context Switches        10002
      Involuntary Context Switches      34
      Block Input Operations            0
      Block Output Operations           124376
```

```
196
197         /*----------------------------------------------------*/
198         /* Step 7: T-Test by Marital Status                  */
199         /*----------------------------------------------------*/
200         proc ttest data=babyweight;
201             class Married;
202             var Weight;
203             title "T-Test: Birth Weight by Marital Status";
204         run;
```

```
NOTE: PROCEDURE TTEST used (Total process time):
      real time           1.47 seconds
      user cpu time       0.99 seconds
      system cpu time     0.20 seconds
      memory              17078.12k
      OS Memory           50832.00k
      Timestamp           04/17/2026 04:33:12 PM
      Step Count                        37  Switch Count  111
      Page Faults                       0
      Page Reclaims                     51830
      Page Swaps                        0
      Voluntary Context Switches        9995
      Involuntary Context Switches      35
      Block Input Operations            0
      Block Output Operations           109360
```

```
205
206         OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
216
```

Results: Assignment 1.sas

Classroom Data: Cooperation vs Competition

Obs	cooperation	competition
1	70	66
2	70	62
3	71	68
4	72	68
5	72	75
6	72	74
7	73	80
8	73	80
9	74	82
10	84	85
11	80	85
12	79	86
13	80	88
14	85	90

Measures of Central Tendency

The MEANS Procedure

Variable	Mean	Median	Mode
cooperation	75.36	73.00	72.00
competition	77.79	80.00	68.00

Measures of Variation

The MEANS Procedure

Variable	Range	Variance	Std Dev
cooperation	15.00	26.71	5.17
competition	28.00	81.10	9.01

Measures of Shape

The MEANS Procedure

Variable	Skewness	Kurtosis
cooperation	0.82	-0.79
competition	-0.40	-1.17

Detailed Univariate Analysis

The UNIVARIATE Procedure
Variable: cooperation

Moments			
N	14	Sum Weights	14
Mean	75.3571429	Sum Observations	1055

Moments			
Std Deviation	5.16805488	Variance	26.7087912
Skewness	0.81813031	Kurtosis	-0.7898024
Uncorrected SS	79849	Corrected SS	347.214286
Coeff Variation	6.8580823	Std Error Mean	1.38122076

Basic Statistical Measures			
Location		Variability	
Mean	75.35714	Std Deviation	5.16805
Median	73.00000	Variance	26.70879
Mode	72.00000	Range	15.00000
		Interquartile Range	8.00000

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	54.55836	Pr > |t|	<.0001
Sign	M	7	Pr >= |M|	0.0001
Signed Rank	S	52.5	Pr >= |S|	0.0001

Quantiles (Definition 5)	
Level	Quantile
100% Max	85
99%	85
95%	85
90%	84
75% Q3	80
50% Median	73
25% Q1	72
10%	70
5%	70
1%	70
0% Min	70

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
70	2	79	12
70	1	80	11
71	3	80	13
72	6	84	10
72	5	85	14

Detailed Univariate Analysis

The UNIVARIATE Procedure
Variable: competition

Moments			
N	14	Sum Weights	14
Mean	77.7857143	Sum Observations	1089
Std Deviation	9.00579789	Variance	81.1043956
Skewness	-0.3969923	Kurtosis	-1.1717322

Moments			
Uncorrected SS	85763	Corrected SS	1054.35714
Coeff Variation	11.5777016	Std Error Mean	2.40690073

Basic Statistical Measures			
Location		Variability	
Mean	77.78571	Std Deviation	9.00580
Median	80.00000	Variance	81.10440
Mode	68.00000	Range	28.00000
		Interquartile Range	17.00000

Note: The mode displayed is the smallest of 3 modes with a count of 2.

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	32.31779	Pr > |t|	<.0001
Sign	M	7	Pr >= |M|	0.0001
Signed Rank	S	52.5	Pr >= |S|	0.0001

Quantiles (Definition 5)	
Level	Quantile
100% Max	90
99%	90
95%	90
90%	88
75% Q3	85
50% Median	80
25% Q1	68
10%	66
5%	62
1%	62
0% Min	62

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
62	2	85	10
66	1	85	11
68	4	86	12
68	3	88	13
74	6	90	14

Sample of Baby Birth Weight Data

Obs	Weight	Black	Married	Boy	MomAge	MomSmoke	CigsPerDay	MomWtGain	Visit	MomEdLevel
1	4111	0	1	1	-3	0	0	-16	1	0
2	3997	0	1	0	1	0	0	2	3	2
3	3572	0	1	1	0	0	0	-3	3	0
4	1956	0	1	1	-1	0	0	-5	3	2
5	3515	0	1	1	-6	0	0	-20	3	0
6	3757	0	1	0	3	0	0	0	3	2

Obs	Weight	Black	Married	Boy	MomAge	MomSmoke	CigsPerDay	MomWtGain	Visit	MomEdLevel
7	2977	1	0	1	-5	1	5	5	3	0
8	3884	0	0	0	-5	0	0	0	3	2
9	3629	0	1	0	6	0	0	-5	3	0
10	3062	0	1	1	-1	0	0	6	3	2

Missing Value Check for Birth Weight

The MEANS Procedure

Analysis Variable : Weight Weight	
N	N Miss
50000	0

Descriptive Statistics for Birth Weight

The MEANS Procedure

Analysis Variable : Weight Weight			
Mean	Std Dev	Minimum	Maximum
3370.76	566.3850556	240.0000000	6350.00

Distribution of Infant Birth Weight

The UNIVARIATE Procedure
Variable: Weight (Weight)

Moments			
N	50000	Sum Weights	50000
Mean	3370.75664	Sum Observations	168537832
Std Deviation	566.385056	Variance	320792.031
Skewness	-0.7554476	Kurtosis	2.69005906
Uncorrected SS	5.84139E11	Corrected SS	1.60393E10
Coeff Variation	16.8029056	Std Error Mean	2.53295097

Basic Statistical Measures			
Location		Variability	
Mean	3370.757	Std Deviation	566.38506
Median	3402.000	Variance	320792
Mode	3402.000	Range	6110
		Interquartile Range	658.00000

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	1330.763	Pr > |t|	<.0001
Sign	M	25000	Pr >= |M|	<.0001
Signed Rank	S	6.2501E8	Pr >= |S|	<.0001

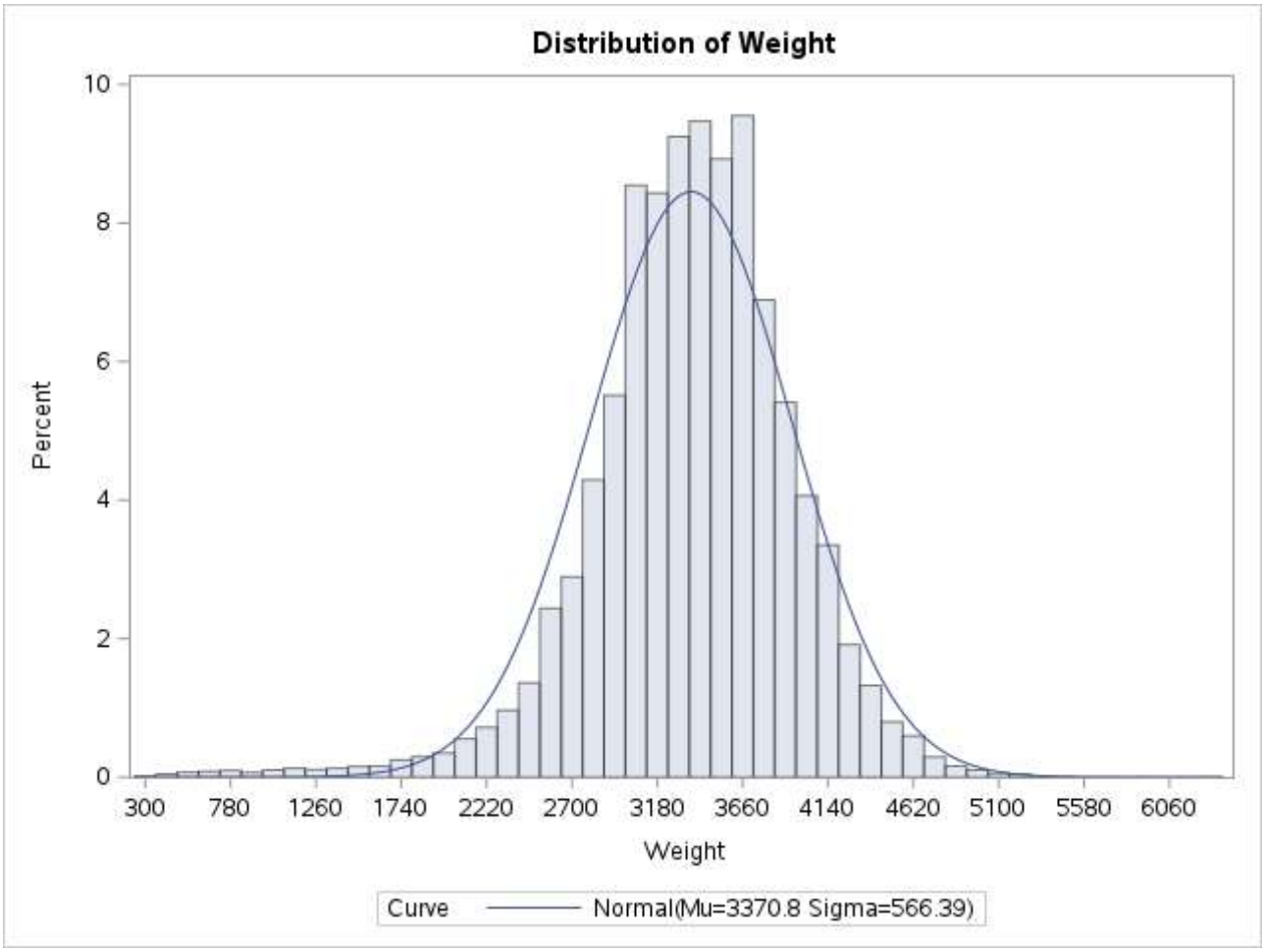
Quantiles (Definition 5)	
Level	Quantile
100% Max	6350

Quantiles (Definition 5)	
Level	Quantile
99%	4605
95%	4224
90%	4026
75% Q3	3720
50% Median	3402
25% Q1	3062
10%	2722
5%	2466
1%	1559
0% Min	240

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
240	45253	5415	44399
284	17687	5642	2262
284	4319	5642	20963
312	18398	5970	25666
322	6789	6350	34693

Distribution of Infant Birth Weight

The UNIVARIATE Procedure



Distribution of Infant Birth Weight

The UNIVARIATE Procedure
Fitted Normal Distribution for Weight (Weight)

Parameters for Normal Distribution		
Parameter	**Symbol**	**Estimate**
Mean	Mu	3370.757
Std Dev	Sigma	566.3851

Goodness-of-Fit Tests for Normal Distribution				
Test	**Statistic**		**p Value**	
Kolmogorov-Smirnov	**D**	0.050153	**Pr > D**	<0.010
Cramer-von Mises	**W-Sq**	33.855093	**Pr > W-Sq**	<0.005
Anderson-Darling	**A-Sq**	232.947399	**Pr > A-Sq**	<0.005

Quantiles for Normal Distribution		
	Quantile	
Percent	**Observed**	**Estimated**
1.0	1559.00	2053.15
5.0	2466.00	2439.14
10.0	2722.00	2644.90
25.0	3062.00	2988.74
50.0	3402.00	3370.76
75.0	3720.00	3752.78
90.0	4026.00	4096.61
95.0	4224.00	4302.38
99.0	4605.00	4688.37

T-Test: Birth Weight by Baby Sex

The TTEST Procedure

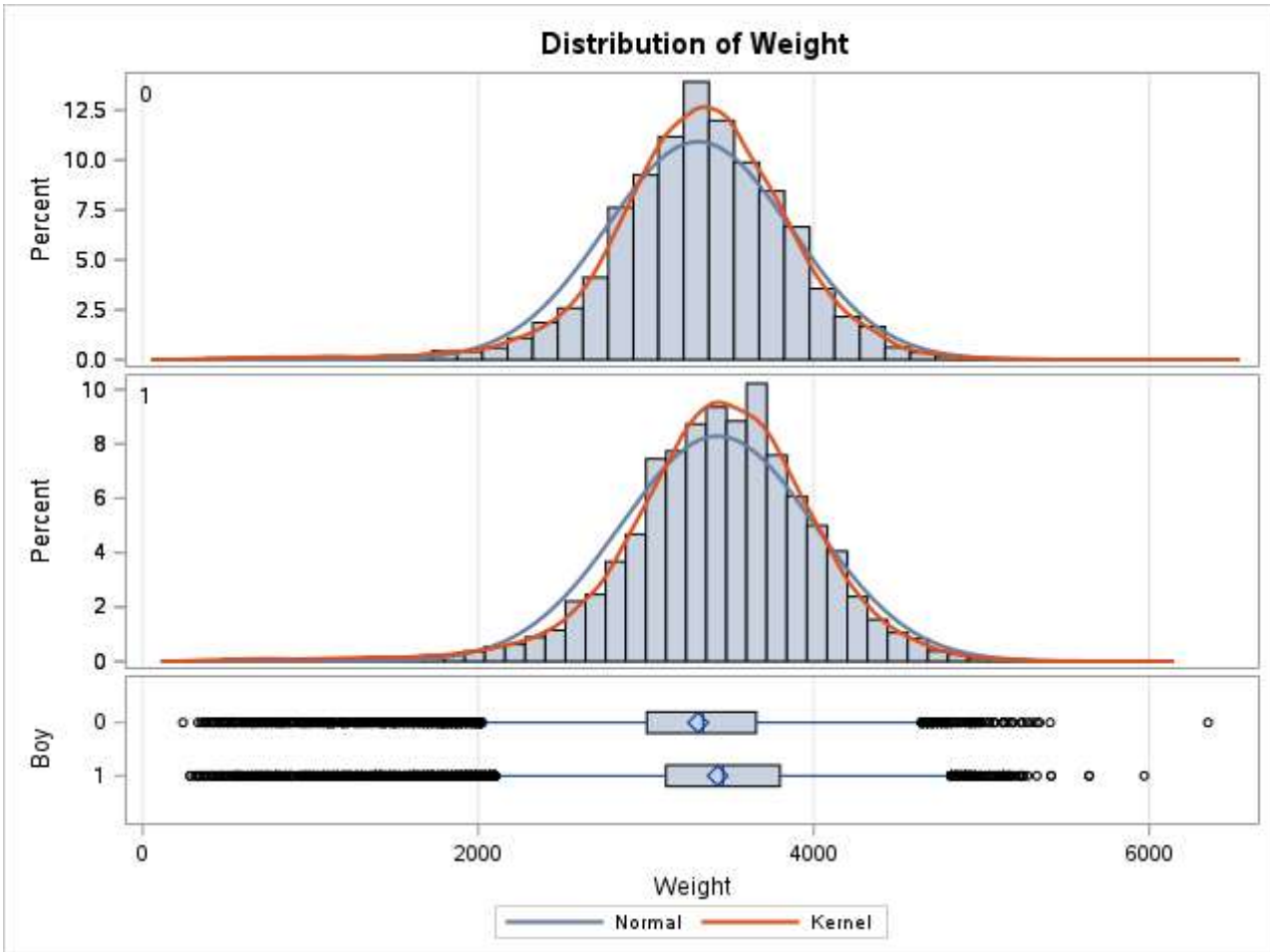
Variable: Weight (Weight)

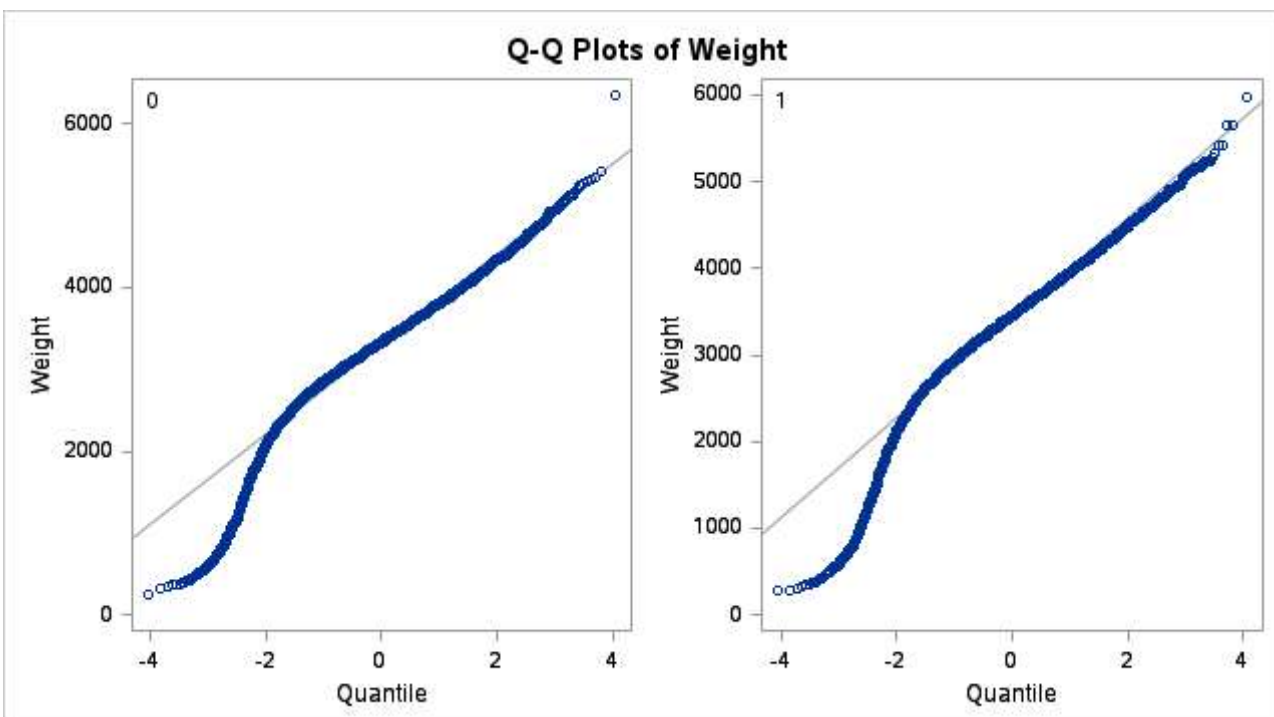
Boy	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
0		24208	3310.6	547.7	3.5204	240.0	6350.0
1		25792	3427.3	577.7	3.5970	284.0	5970.0
Diff (1-2)	**Pooled**		-116.7	563.4	5.0416		
Diff (1-2)	**Satterthwaite**		-116.7		5.0331		

Boy	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
0		3310.6	3303.7	3317.5	547.7	542.9	552.7
1		3427.3	3420.2	3434.3	577.7	572.7	582.7
Diff (1-2)	**Pooled**	-116.7	-126.6	-106.8	563.4	559.9	566.9
Diff (1-2)	**Satterthwaite**	-116.7	-126.6	-106.8			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	49998	-23.15	<.0001
Satterthwaite	Unequal	49993	-23.18	<.0001

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	25791	24207	1.11	<.0001





T-Test: Birth Weight by Maternal Smoking

The TTEST Procedure

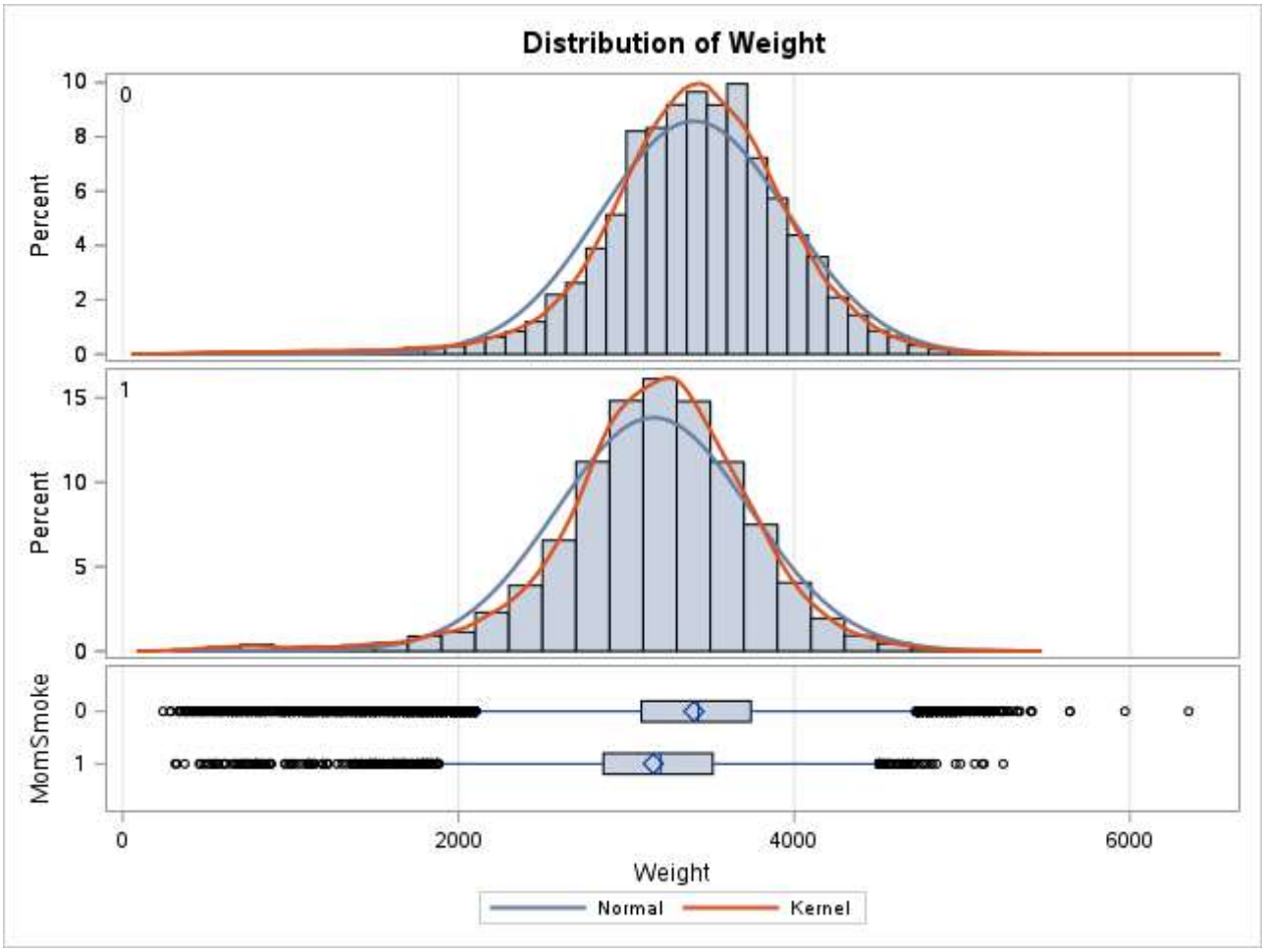
Variable: Weight (Weight)

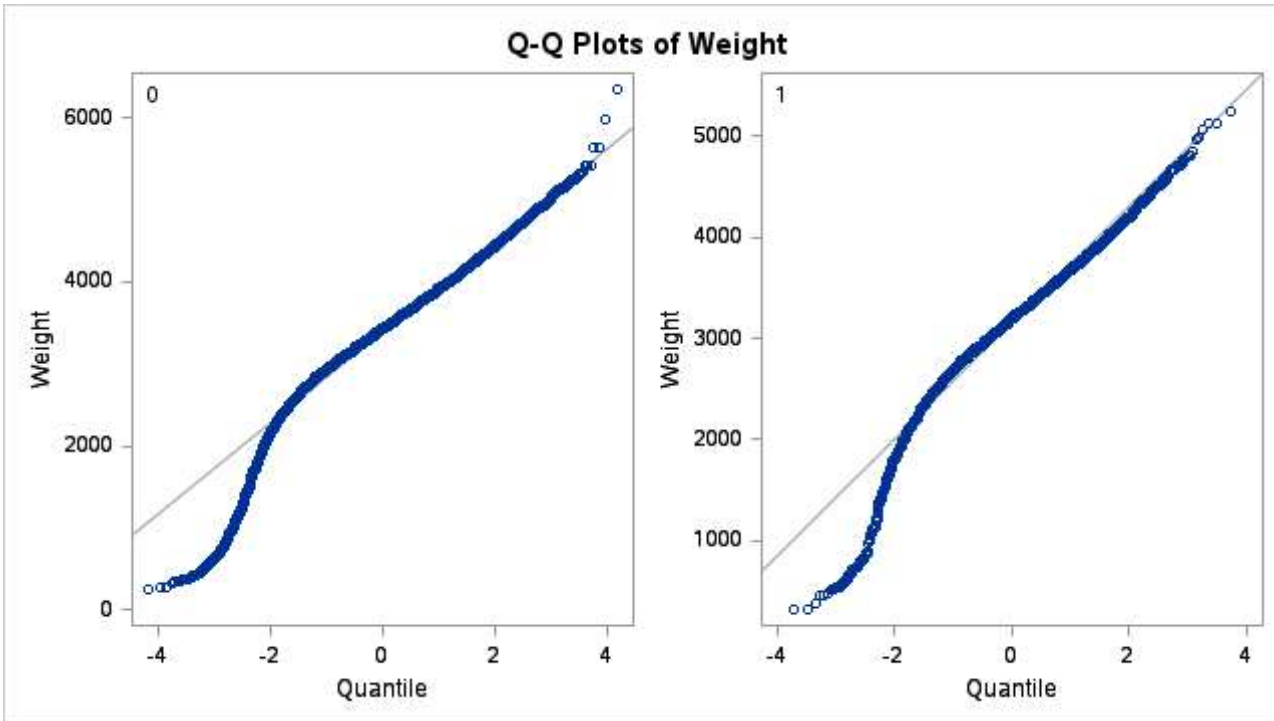
MomSmoke	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
0		43467	3402.3	558.0	2.6766	240.0	6350.0
1		6533	3160.9	576.8	7.1358	312.0	5245.0
Diff (1-2)	Pooled		241.5	560.5	7.4376		
Diff (1-2)	Satterthwaite		241.5		7.6213		

MomSmoke	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
0		3402.3	3397.1	3407.6	558.0	554.3	561.8
1		3160.9	3146.9	3174.8	576.8	567.0	586.8
Diff (1-2)	Pooled	241.5	226.9	256.0	560.5	557.1	564.0
Diff (1-2)	Satterthwaite	241.5	226.5	256.4			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	49998	32.46	<.0001
Satterthwaite	Unequal	8474.1	31.68	<.0001

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	6532	43466	1.07	0.0004





T-Test: Birth Weight by Marital Status

The TTEST Procedure

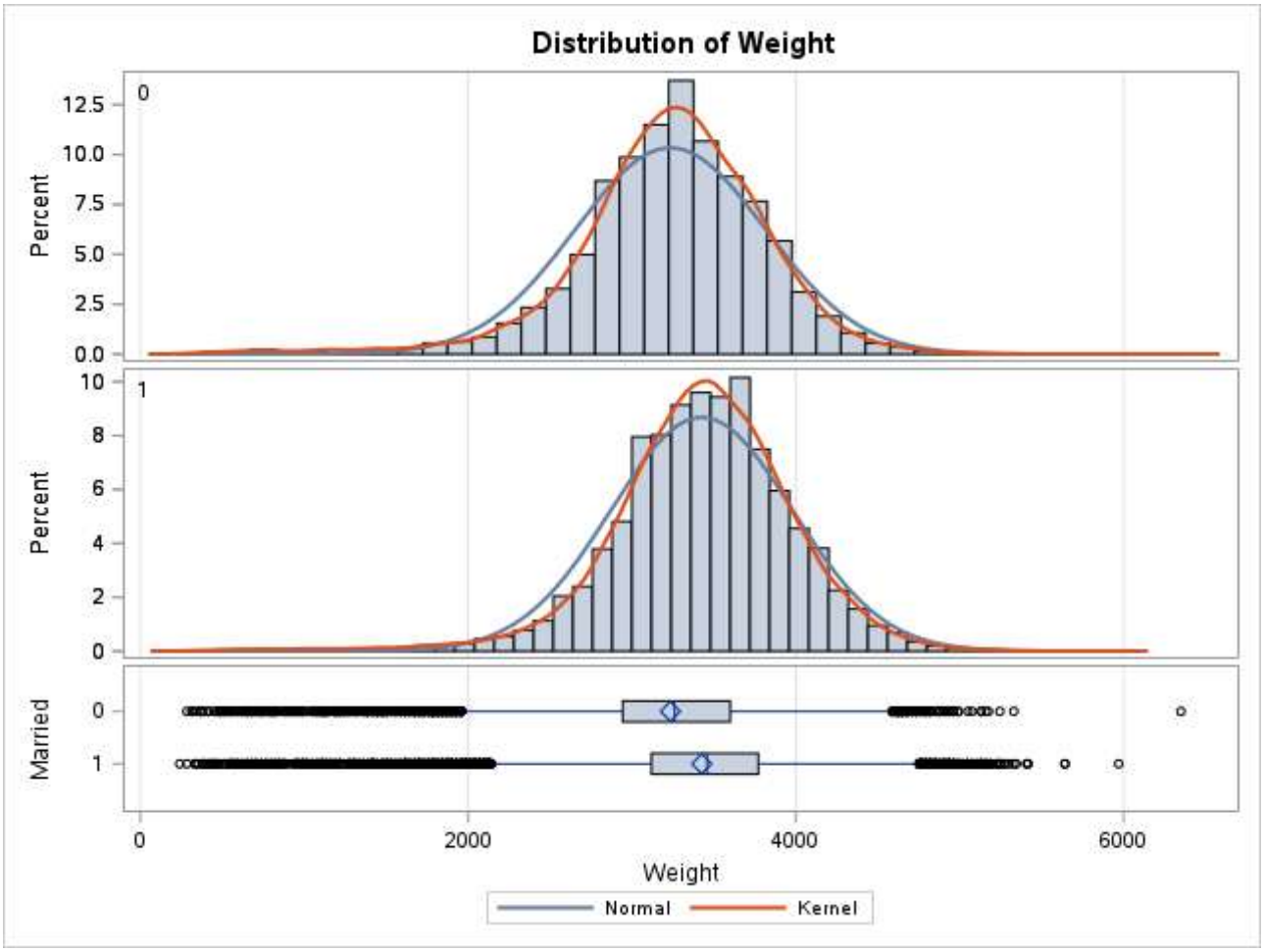
Variable: Weight (Weight)

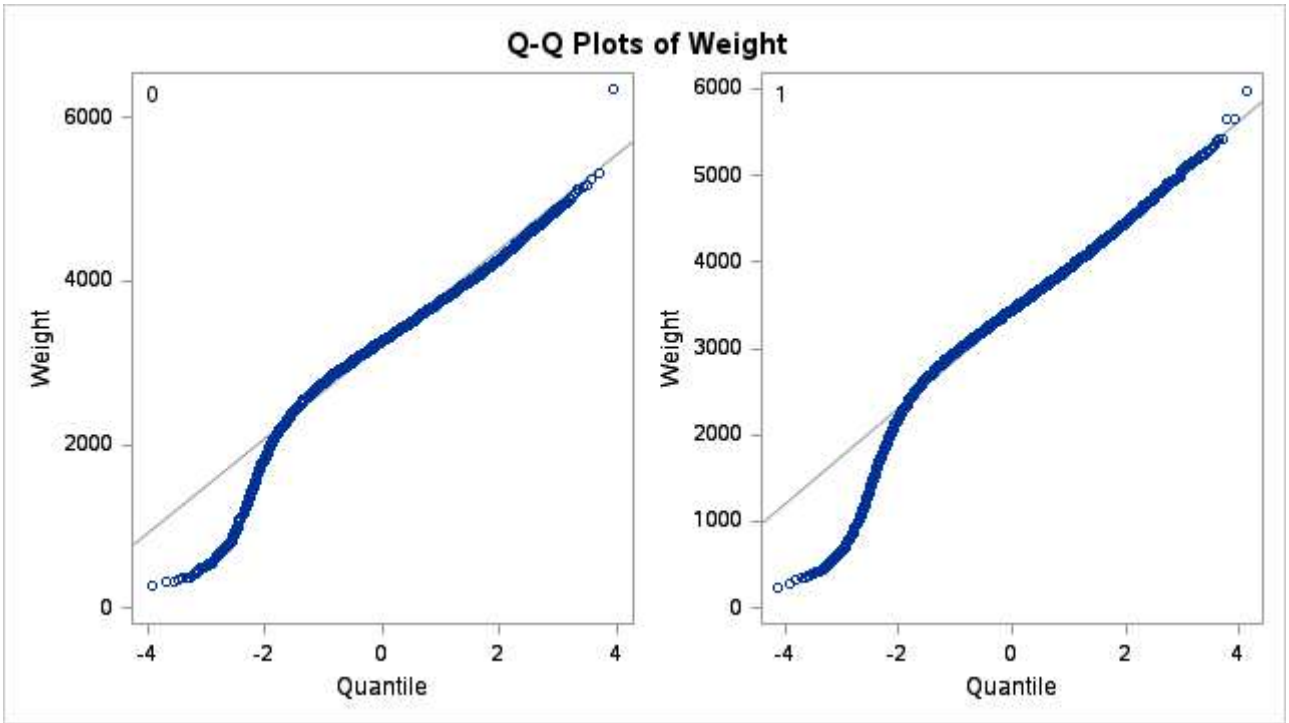
Married	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
0		14369	3234.4	579.0	4.8302	284.0	6350.0
1		35631	3425.7	551.8	2.9231	240.0	5970.0
Diff (1-2)	Pooled		-191.3	559.7	5.5315		
Diff (1-2)	Satterthwaite		-191.3		5.6459		

Married	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
0		3234.4	3225.0	3243.9	579.0	572.4	585.8
1		3425.7	3420.0	3431.5	551.8	547.8	555.9
Diff (1-2)	Pooled	-191.3	-202.1	-180.5	559.7	556.3	563.2
Diff (1-2)	Satterthwaite	-191.3	-202.4	-180.2			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	49998	-34.58	<.0001
Satterthwaite	Unequal	25443	-33.88	<.0001

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	14368	35630	1.10	<.0001


Distribution of Weight
0
1
Percent
Married
Weight
Normal
Kernel


Q-Q Plots of Weight
0
1
Weight
Quantile